

The technology

Computers use digital data comprising of 1's and 0's. Sound on the other hand is an analog signal with smooth curves and varying frequencies brought together in a wondrous melody. So in order for digital data (which was



all that could be found) to be converted into an analog signal we needed special encoders. So the basic sound card consisted of an ADC(Analog-to-Digital) convertor for microphones to connect to and convert the analog sound into a digital stream that could be stored within a computer. And a DAC(Digital-to-Analog) convertor which did the exact opposite. Since then the technology has significantly advanced and we now use single microchips called Codecs which incorporate the functions of both the DAC and the ADC. Conversion of an analog signal to digital does lead to data loss as digital signals are basically a step



function, so in order to reproduce the analog signal accurately the sound card needs to sample the analog signal at a high frequency. Thus, the higher the sampling rate, the better recording you can get done on that particular sound card. Then again there are DSP (Digital Signal Processors) which are situated between the DAC and ADC stage and are capable of performing heavy duty calculations. These come with their own memory too. This way the processing of the sound signal is isolated completely from the computer processor. Otherwise, the job of the DSP is done by the CPU which puts unnecessary load on it.

SSD

The latest entrant in the storage media evolution is the SSD or Solid State Drive. You've read about SSDs and you've read about RAM. One is faster than the other while the other is non-volatile so the logical step was to create a drive that would be faster and non-volatile. Thus, the SSDs were born. Now both RAM chips and Hard drives have been around for quite a long time but it